

Problem Analysis Report: Urban Municipal Solid Waste Management

Name: Ankita Mukund Panpatil

Deliverable: Module 2 – Problem Analysis Report

1. Environmental System Overview

The selected environmental system is the **Urban Municipal Solid Waste (MSW) Collection and Processing System**. This system manages household and commercial waste through stages of generation, segregation, municipal transport, mechanical/manual sorting at recovery facilities, and final disposal or recycling loops.

2. Problem Statement

The primary challenge within this system is the **severe inefficiency of mixed-waste processing driven by low source-segregation rates**. Because organic and inorganic waste streams are predominantly combined at the household level, recyclable materials become contaminated, degrading their commercial value and overwhelming sorting infrastructure, which ultimately leads to excessive landfill accumulation and environmental pollution.

3. Root Causes

- **Lack of Civic Awareness & Training:** Insufficient public education regarding the long-term environmental impacts of improper waste disposal and unclear household sorting guidelines.
- **Inadequate Infrastructure:** Absence of mandatory dual-bin infrastructure in residential zones, making proper sorting inconvenient for residents.
- **Rigid Municipal Logistics:** Outdated, fixed-schedule truck routing that fails to account for actual waste accumulation levels across different neighborhoods.

4. System Gaps

System Area	Identified Gap
Source Separation	No enforceable incentives or penalties for households regarding waste segregation.
Recovery Facilities	Mechanical sorting units lack advanced optical sorters for high-speed plastic identification, resulting in high manual labor dependency.
Data & Monitoring	Absence of real-time IoT tracking for bin fill levels and fleet efficiency.

5. Potential Impact

- **Environmental Impact:** High volumes of organic waste decomposing in landfills generate massive methane emissions and toxic leachate that contaminates local groundwater tables.
- **Economic Impact:** High municipal expenditures on sorting labor and fuel due to inefficient transport routes and low material recovery values.
- **Public Health & Operational Impact:** Overflowing community bins attract disease vectors and create severe sanitary hazards for sanitation workers and nearby residents.